Supplementary Information for Rethinking News Framing with Large Language Models

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1 Materials and Methods

1.1 Event clustering

To facilitate the mixing of statements from different articles, it's essential that all articles discuss the same issue. Otherwise, the LLM struggles to construct a coherent narrative from the sampled sentences. Therefore, we grouped articles addressing the same topic around the same date. The restriction on publication dates accounts for evolving issues, ensuring that all statements are meaningfully related. To achieve this, we organized articles using a moving three-day window and clustered them based on semantic similarity.

For clustering, we used GPT-4o's large-context embedding model to generate embeddings for each article, performing pairwise cosine similarity comparisons between articles. We then constructed a graph where nodes represent individual articles, and edges indicate the cosine similarity of their embeddings. By removing edges below an empirically determined threshold of 0.8, we isolated densely connected components representing cohesive clusters of articles focused on specific news events. To enhance interpretability, GPT-4 was employed to assign thematic titles to these clusters based on the article titles within each group. This approach resulted in 16,453 unique clusters with at least two articles, spanning from January to September 2024.

1.2 Event sampling

Table S1 presents the events selected for the experiment, along with the associated cluster size (number of articles published about each event) and the corresponding event dates.

A critical consideration in the sampling process was ensuring a balanced representation of sentence types and tones, which is essential for generating multiple versions of the

same story. Figure S1 illustrates the distribution of sentence types (facts, quotes, opinions) and tones (positive, neutral, negative) across the seven selected events. As observed, opinions are used much less frequently compared to facts and quotes. Additionally, neutral sentences are more common, with an imbalance often present between positive and negative facts, depending on the event's nature. For instance, in more positive events like the Bipartisan Tax Bill, there are more positive facts, while for the Boeing 737 Grounding, negative facts prevail. Note that sentences were also classified as "borderline" and "other," but these were rare, so our focus remained on the primary types.

Nevertheless, Figure S1 demonstrates that the selected events are sufficiently balanced, allowing the ensemble of articles to provide enough material to create narratives with a similar tone across sentences. The full list of selected sentences is available on the project's OSF page: `https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe`.

Event	Cluster Size	Date
Trump Assassination Attempt	140	2024-07-13
Boeing 737 Max 9 Grounding	31	2024-01-07
Fed Interest Rate Cut	15	2024-09-18
US-Russia Prisoner Swap	25	2024-08-02
Bipartisan Tax Bill Passed	8	2024-01-29
Imane Khelif's Olympic Journey	8	2024-08-06
Harris Picks Tim Walz	96	2024-08-06

Table S1: The list of events selected for our experiment. Cluster size indicates the number of articles published by all publishers about that event.

1.3 Stimuli generation prompt

For generating the synthetic articles, the objective was to remain as faithful as possible to the original content from actual news articles. This approach demonstrates how the selection of information can influence perceptions and ensures that the LLM does not alter the reported facts. To achieve this, we utilized the model's temperature parameter, which controls the randomness and creativity of the output. By setting the temperature to zero, we ensured that input sentences remained unchanged in the output, preserving their original

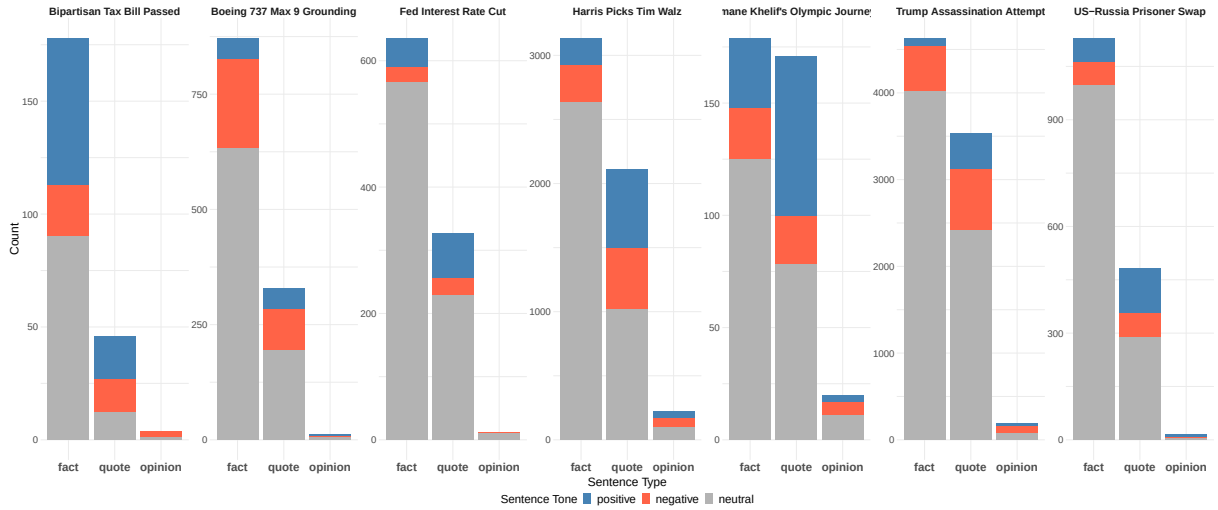


Figure S1: Distribution of sentence types (facts, quotes, opinions) and tones (positive, neutral, negative) across the seven selected events. Note that the scales vary, as the total number of sentences for each event depends on the cluster size.

form. We used GPT-4 for all tasks, and the top_p parameter was set to 0. Figure S2 shows the article generation prompt. The generated stimuli are available on the project's OSF page: `https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe`.

1.4 Descriptive analysis of stimuli

In this section, we conduct a descriptive analysis of the generated stimuli to better understand how each version of each event portrays the story. We provided the three versions of each topic to a GPT-4 model and asked it to analyze the differences and summarize the framing of each article. The prompt for the analysis can be found in Figure S3. Figure S4 displays the framings identified by the LLM. The results demonstrate that a coherent frame emerges for each event, resulting from clustering sentences based on their tone. Notably, we did not explicitly define the framing in each condition; these are byproducts of the pipeline's design.

1.5 Negativity bias

In this section, we provide evidence for the prevalence of negativity bias in the media. We utilized the dataset from [1], which compiles the top 20 news articles from the homepages

Article Generation Prompt

```
You are a journalist who writes news articles. Read the
following statements and perform the task that follows:

{list_of_20_sentences}

You are provided with a list of factual statements,
opinions, and quotes. Based on these statements, write
a news article about {event_theme}. Use all the provided
statements. Include a proper title for the article.
Write the article with a {tone}.
```

Figure S2: The article generation prompt. The placeholder variables in red are parametrized by passing in the relevant information at those positions.

of ten prominent mainstream media outlets across the political spectrum, spanning January to September 2024. During this period, the dataset contains 72,132 unique articles and 16,453 events with at least two articles. Note that these events are defined based on a rolling three-day window, so article overlap is possible.

For each event, we calculated the average coverage tone by averaging the tone of the corresponding articles. Figure S5 displays the distribution of tone across all events. Evidently, most news has a negative tone, as shown by the large mass on the negative side of the spectrum.

The figure also highlights the tone of the events we sampled for our experiment. A side effect of this negativity bias is that, for most major events, there are very few positive facts, quotes, or opinions. Consequently, we had to over-sample positive events to ensure we could meet the strict balances we set for sentence types and tones.

1.6 Regression analysis

Our main analysis in the paper estimates the following equation using OLS.

$$Y_{it} = \beta_0 + \beta_1 \mathrm{P}_{it} + \beta_2 \mathrm{N}_{it} + \mathbf{X}_{it}\mathbf{\Gamma} + \theta_t + \epsilon_{it} \tag{1}$$

where:

- Y_{it} represents the outcome variable (either feeling towards the subject of the article,

Framing Analysis Prompt

```
You are given three different versions of the same news
story, each with a particular framing that emphasizes
certain aspects of the event.  Your task is to analyze
the differences in framing between these versions and
generate a short one-sentence summary that best describes
the distinct framing of each article.  The three versions
are as follows:
Article 1:  {positive article}
Article 2:  {neutral article}
Article 3:  {negative article}
```

Figure S3: The prompt for descriptive framing analysis of the stimuli. The placeholder variables in red are parametrized by passing in the relevant information at those positions.

or the first principal component of opinion questions) for participant i on topic t.

- P_{it} is a dummy variable for whether the article seen by participant i on topic t is has a positive framing.
- N_{it} is a dummy variable for for whether the article seen by participant i on topic t has a negative framing.
- $\mathbf{X}_{it}\mathbf{\Gamma}$ represents the vector of demographic covariates (partisanship, ideology, gender, education, race, age) and media consumption habits.
- $\theta_{t(i)}$ represents topic fixed effects.
- ϵ_{it} is the error term.

1.7 Survey questions

This sections outlines the full list of outcome measures used in the experiment. To systematically generate these questions, we input all the facts from the article into GPT-4o and requested a list of relevant questions. Researchers then selected the six questions that were most appropriate and relevant to the facts presented in the articles. For the full survey refer to the project's OSF page: `https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe`.

Generic Questions for All Events

1. Prior to reading the article, how informed were you about the topic of the article?
2. Regardless of your personal feelings regarding the topic of the article: how do you rate the tone of the article overall?
3. Regardless of what you already knew about the topic of the article: how informative do you think the article was about the topic it covered?
4. How do you rate your feelings towards the main subject of the article?

Trump Assassination Attempt

Event Description: Assassination Attempt on Donald Trump at Pennsylvania Rally Sparks Political Turmoil and Security Concerns Ahead of 2024 Republican National Convention

1. How would you rate Donald Trump's chances of being re-elected as president in the upcoming election?
2. How satisfied are you with Democrats' response to this assassination event?
3. How do you perceive the safety at campaign rallies for high-profile political candidates?
4. How much sympathy do you feel for Donald Trump?
5. How much trust do you have in the Secret Service and security measures in place to protect presidential candidates?
6. To what extent do you think this event would impact the political landscape for the upcoming 2024 election?

US-Russia Prisoner Swap

Event Description: Historic US-Russia Prisoner Swap Brings Americans Home

1. How do you perceive the success of diplomatic efforts based on the recent U.S.-Russia prisoner swap?

2. How confident are you in the U.S. government's ability to secure the release of American detainees abroad?

3. How strongly do you believe that prisoner exchanges encourage further hostage-taking by hostile regimes?

4. How has the recent prisoner swap impacted your perception of the Biden administration?

5. How fair do you believe international negotiations are between the U.S. and hostile nations?

6. How do you perceive U.S. leadership in global affairs following the recent prisoner swap?

Bipartisan Tax Bill Passed

Event Description: House Passes Bipartisan Tax Bill Expanding Child Tax Credit and Business Tax Breaks

1. How do you perceive the recent tax bill passed by the House of Representatives?

2. How beneficial do you believe the increase in the child tax credit will be for American families?

3. How supportive are you of providing substantial tax breaks for businesses?

4. How supportive are you of the expansion of the child tax credit as described in the recent legislation?

5. How well do you believe the recent tax bill supports low-income families?

6. How do you think the recent tax bill will affect economic inequality in the United States?

Imane Khalif's Olympic Journey

Event Description: Imane Khelif: Algerian Boxer Defies Gender Misconceptions on Road to Olympic Glory

1. How has your perception of women's boxing as a competitive sport changed?
2. How concerned are you about gender misconceptions and scrutiny in sports?
3. How supportive are you of female athletes who face gender eligibility controversies?
4. How has your confidence in the fairness and integrity of Olympic sports been affected?
5. To what extent do you believe that athletes like Imane Khelif inspire others and break down societal barriers?
6. How likely are you to follow women's boxing and support its athletes in the future?

Harris Picks Tim Walz

Event Description: Kamala Harris Selects Tim Walz as Vice Presidential Running Mate

1. How do you perceive Kamala Harris's decision to select Tim Walz as her running mate for the upcoming election?
2. How confident are you in the ability of the Harris-Walz ticket to address the needs and concerns of the American people?
3. How confident are you in Tim Walz's ability to fulfill the responsibilities of the vice presidency?
4. How well do you believe the Harris-Walz ticket embodies progressive values such as support for unions, codifying abortion rights, and common-sense gun control?
5. How likely do you think the Harris-Walz ticket will be able to energize young voters and independents in swing states?
6. How likely are you to support the Democratic Party's strategy based on the selection of Tim Walz as running mate?

Boeing 737 Max 9 Grounding

Event Description: Boeing 737 Max 9 Planes Grounded After Alaska Airlines Incident Due to Fuselage Blowout and Loose Parts

1. How do you perceive Boeing's commitment to ensuring the safety of its aircraft?
2. How safe do you feel flying on commercial airlines?
3. How comfortable would you feel flying on a Boeing 737 Max aircraft in the future?
4. How confident are you in the professionalism and capabilities of airline crews to handle in-flight emergencies?
5. How confident are you in the FAA's oversight of airline safety?
6. How much do you support the immediate grounding and inspection of aircraft when a serious safety issue is identified?

Fed Interest Rate Cut

Event Description: U.S. Inflation Eases for the First Time in 2024, Offering Relief to Consumers and Markets

1. How do you perceive the Federal Reserve's decision to cut interest rates?
2. How confident are you in the Federal Reserve's independence from political influence?
3. How would you rate economic conditions in the United States today?
4. How stable do you believe the U.S. economy is after the rate cut?
5. How appropriate do you believe the timing of the Federal Reserve's rate cut is?
6. How likely do you think the rate cut will provide long-term economic benefits?

Table S2: Demographic characteristics of survey respondents (N = 2,141)

Characteristic	n (%)
Gender	
Female	1,049 (49%)
Male	1,077 (50%)
Other (please specify)	15 (0.7%)
Age	
18–24 years	29 (1.4%)
25–34 years	376 (18%)
35–44 years	626 (29%)
45–54 years	495 (23%)
55–64 years	350 (16%)
65+ years	265 (12%)
Education	
Some high school or less	13 (0.6%)
High school diploma or GED	178 (8.3%)
Some college, no degree	321 (15%)
Associate or technical degree	244 (11%)
Bachelor's degree	953 (45%)
Graduate or professional degree	426 (20%)
Prefer not to say	6 (0.3%)
Race/Ethnicity	
American Indian or Alaska Native	15 (0.7%)
Asian	303 (14%)
Black or African American	125 (5.8%)
White or Caucasian	1,572 (73%)
Other	34 (1.6%)
Prefer not to say	16 (0.7%)
Multiracial combinations[1]	See note below
Party Identification	
Democrat	942 (44%)
Independent	607 (28%)
Republican	546 (26%)
Other (please specify)	46 (2.1%)

[1] Multiracial responses include combinations such as White and Asian (1.0%), White and Black (0.6%), and other categories (each ¡1%). Full breakdown available upon request.

1.8 Respondents' demographics

We recruited a total of 2,141 respondents through Amazon Mechanical Turk (MTurk). The sample included a high-quality panel of participants with whom we have worked in previous studies and whose reliability and attentiveness have been verified. Table S2 provides summary statistics on key demographic characteristics of the respondent pool.

1.9 Article length comparison across conditions

To assess the consistency of the experimental stimuli, we examined differences in article length across treatment conditions. Figure S3 reports the average word counts with 95% confidence intervals for negative, neutral, and positive articles. On average, negative articles contained 540 words, neutral articles 601 words, and positive articles 610 words. These differences were not statistically significant when averaged across all topics.

The observed variation reflects the fact that the original wording of sentences from real news articles was preserved rather than rewritten to standardize length. While this introduces some natural variability, the number of sentences was fixed across conditions, ensuring that each article contained the same number of factual and quoted statements. This design choice prioritized fidelity to the source material while maintaining structural consistency across experimental conditions.

1.10 Example Stimuli

To provide better clarity regarding the experimental stimuli, we include here the full set of synthetic articles used for one example event: *Federal Reserve Interest Rate Cuts*. The three articles below correspond to the Negative, Neutral, and Positive framing conditions presented to respondents in the experiment. The full list of synthetic articles is publicly available on the project's OSF page: `https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe`.

Negative Framing Condition

Federal Reserve's Rate Cut Sparks Controversy and Concerns of Political Motives

In a move that has ignited a firestorm of criticism and speculation, the Federal Reserve has decided to lower interest rates by half a percentage point, a decision that has been met with skepticism and accusations of political interference. The timing of the rate cut, just weeks before the 2024 presidential election, has drawn ire from various quarters, including former President Donald J. Trump, who is the Republican candidate in the upcoming election.

House Ways and Means Committee Chairman Jason Smith, R-Mo., expressed his disapproval, stating that the Federal Reserve is "finally" lowering interest rates, "even though prices are still 20 percent higher, paychecks are worth less, and more families are taking on credit card debt." He further criticized the current administration, saying, "Americans aren't just looking for relief from high interest rates, they're looking for relief from the same old failed Biden-Harris policies that crushed them to begin with."

Some Republicans have argued that the Fed's decision to cut rates is a strategic move to benefit Democrats. Sen. Kevin Cramer, R-N.D., voiced his concerns on social media, writing, "The Federal Reserve claims political independence." This sentiment was echoed by Trump, who suggested, without evidence, that the central bank's actions were politically motivated. "I guess it shows the economy is very bad, to cut it by that much – assuming they're not just playing politics," Trump remarked during a stop at a Bitcoin bar in New York City.

The rate cut has also drawn criticism from within the Democratic Party. Senator Elizabeth Warren has harshly criticized Fed Chair Jerome Powell for his previous aggressive rate hikes and the delayed pivot to cuts. "The Fed makes it more likely that the U.S. will end up in recession every day that they delay, or if they do just a small rate [cut] that indicates they're trying to keep a tight rein on an economy that doesn't need that kind of restraint," Warren told HuffPost.

Markets initially reacted positively to the news but closed in the red, partly due to fears that the Fed's actions were an attempt to preempt an unwanted economic slowdown. Critics worry that the rate cut could inadvertently speed up the economy, causing inflation to remain at an uncomfortably high level. Others fear that the delay in rate cuts could lead to job losses and a rise in the unemployment rate, which is expected to climb to 4.4% by the end of the year.

Vice President Kamala Harris has countered Trump's criticisms by highlighting the potential negative impact of his proposed tariffs on all imports, which she argues would further raise prices for consumers. Meanwhile, Vance, Trump's running mate, dismissed the significance of the rate cut, stating, "A half-a-point is nothing compared to what American families have been dealing with for the last three years."

As the debate rages on, the American public remains caught in the crossfire, grappling with record credit card debt and high delinquency rates. The Federal Reserve's decision, whether politically motivated or not, has left many questioning the soundness of its decision-making and the true state of the U.S. economy.

Neutral Framing Condition

Federal Reserve Cuts Interest Rates for Economic Boost

In a decisive move, the Federal Reserve cut interest rates by half a percentage point on Wednesday, marking an unusually large adjustment. This decision signals that central bankers believe they are making progress in their battle against inflation and are now shifting their focus to safeguarding the job market. The rate cut, which was anticipated by nearly every forecaster, reflects a growing confidence that inflation is moving sustainably toward the Fed's 2% target.

While Fed votes often garner unanimous support, this decision saw dissent from one policymaker, Governor Michelle Bowman, who preferred a more modest quarter-point cut. The probability of a 50-basis-point cut had risen significantly, from 25% a month ago to 64

Officials have been closely monitoring a recent uptick in the unemployment rate, which rose from 3.7% in January to 4.2

Fed Chair Jerome Powell emphasized during his news conference that the central bank is prepared to adjust the pace of rate cuts if the economy proves to be weaker or stronger than expected. This cautious approach is reinforced by economic projections suggesting a more measured pace of rate cuts than officials had envisioned just a few months ago.

The Fed's policy statement indicated that the decision was based on "greater confidence that inflation is moving sustainably toward 2%," and that the central bank "judges that the risks to achieving its employment and inflation goals are roughly in balance." As inflation has declined and the labor market has cooled, the upside risks to inflation have diminished, while the downside risks to unemployment have increased.

Officials now anticipate making another half-point reduction before the end of the year. Policymakers must still decide how much and how quickly to lower interest rates in the coming months and years to achieve their goals. The September decision is seen as a preemptive strike to increase the odds of achieving a "soft landing," where high rates bring down inflation without causing a recession.

Rebecca Patterson, former chief investment strategist at Bridgewater Associates, cautioned that just as Fed officials and private-sector economists misjudged the impact of higher rates on the economy two years ago, "there's a risk the easing cycle will see some similar challenges."

The Fed's decision comes as the average new credit card rate remained unchanged at 24.92% in September, the highest figure since 2019, when LendingTree began tracking the data. This move is part of the Fed's ongoing effort to strike a careful balance between slowing growth enough to ensure price stability without cooling the economy so much that unemployment soars and a recession ensues.

The Fed had previously slashed rates to near zero in March 2020 as the Covid-19 pandemic brought global commerce to a standstill, maintaining those levels for two years. As the economy continues to navigate post-pandemic challenges, the central bank's latest rate cut reflects its commitment to fostering economic stability and growth.

Positive Framing Condition

Federal Reserve Cuts Interest Rates for Economic Boost: A Positive Shift for the U.S. Economy

In a significant move that marks a major shift in monetary policy, the Federal Reserve cut interest rates by a half-point on Wednesday, signaling a preliminary victory in its battle against high inflation. This decision, the first rate cut in four years, has been met with optimism from both markets and households, as it promises to bring much-needed relief from the era of dangerously high borrowing costs.

The resilience of the U.S. economy has made Federal Reserve officials hopeful that they might achieve a historically rare "soft landing," where the economy is steered onto a healthy and sustainable track without triggering a recession. "The U.S. economy is in a good place, and our decision today is designed to keep it there," said Federal Reserve Chairman Jerome Powell. "It's growing at a solid pace, inflation is coming down, the labor market is in a strong place."

Stocks responded favorably to the Fed's rate cut decision, with both the Dow Jones Industrial Average and the S&P 500 hitting all-time highs in the immediate aftermath of the announcement. Anticipation of the rate cuts had already buoyed Wall Street earlier in the week, with the Dow Jones Industrial Average reaching a new record on Monday.

The move is expected to ripple through the economy, providing the first dose of relief in years to Americans who have struggled with high borrowing costs for mortgages, credit cards, and auto and other loans. Homeowners will be able to refinance mortgages at lower rates, saving on monthly payments, and even shift credit card debt to lower-cost personal loans or home equity lines. Nevada real estate agent Zoila Sanchez expressed optimism, telling CNN's John King that lower interest rates would bring welcome relief to residents looking to buy a house or refinance.

"The labor market is in solid condition, and our intention with our policy move today is to keep it there," Powell emphasized. The Fed's decision reflects confidence that inflation is moving sustainably toward the 2 percent target, with Powell noting, "Inflation is now much closer to our objective, and we have gained greater confidence that inflation is coming down."

The rate cuts should, over time, lead to lower borrowing costs for mortgages, auto loans, and credit cards, boosting Americans' finances and supporting more spending and growth. This positive outlook is further supported by plans to lower health-care, housing, and grocery costs, which aim to alleviate everyday expenses that remain higher than pre-pandemic levels.

In fact, the current economic landscape is arguably better than it was in 1995, the last time the Fed convincingly achieved a soft landing. "We made a good, strong start to this, and that's really, frankly, a sign of our confidence — confidence that inflation is coming down," Powell stated.

As the Federal Reserve turns the page on high inflation, this rate cut marks a hopeful step toward sustained economic growth and stability, offering a brighter financial future for households and businesses alike.

2 Experiment Results

2.1 Comprehension check

To minimize experimenter demand effects and avoid any form of deception, participants were presented with the following instructions prior to reading the main article. The instructions explicitly stated that "the article presents just one perspective on the topic, and that other perspectives might be equally consistent with the facts." Following this, participants were asked three comprehension questions to ensure they had read the instructions carefully.

> **Please read these instructions carefully. We will ask you about them on the next page.**
>
> In this survey, we will show you an article about some topic of contemporary interest such as the economy or crime. Please read the article carefully, paying attention to the facts that are mentioned. However, bear in mind that the article presents just one perspective on the topic, and that other perspectives might be equally consistent with the facts.
>
> Once you have read the article, we will ask you some questions. These questions will not be about the details of the article but instead about the general topic.
>
> You do not need to base your answers on the article alone. Rather, use your general knowledge of the topic, including other news articles you may have read.

After reading the instructions, participants were asked three comprehension questions to ensure they understood the material. Of the 2,141 participants, 1,442 answered all questions correctly, 468 made one error, 182 made two errors, and 49 answered all three questions incorrectly. As pre-registered, we excluded from the main analysis any participants who answered more than one instruction check question incorrectly, resulting in a final sample of 1,910 responses. The main findings are robust to this exclusion criterion and remain qualitatively consistent even when applying a stricter threshold that excludes all participants with any incorrect answer, or including everyone regardless of their attention level. See Section 2.5 for detailed robustness checks.

2.2 Moderation by prior informedness

One of the post-treatment questions asked participants to self-assess how informed they were about the subject of the article. While this approach avoids priming participants before reading the article, it introduces a methodological concern: because informedness was measured after the treatment (i.e., article exposure), it may have been influenced by the treatment itself. This poses a challenge for causal inference, as conditioning on post-treatment variables can bias moderation analyses.

Nevertheless, we believe the moderation analysis still yields meaningful insights under a set of reasonable assumptions:

- **Assumption 1 (Monotonic Response):** Exposure to the article can only increase a participant's self-reported informedness. That is, someone who was previously low-informed might come to believe they are high-informed after reading the article, but the reverse (a previously high-informed person rating themselves as low-informed) is unlikely.

- **Assumption 2 (Non-differential Response Bias):** The effect of the treatment on self-reported informedness is independent of its effect on the outcome variables (i.e., feelings and opinions). In other words, individuals who shift their self-assessment from low- to high-informed due to the article do not systematically respond differently to the main outcome measures, relative to participants who did not change their informedness self-assessment.

If these assumptions hold, the estimated treatment effect for the self-identified low-informed group remains unbiased. The estimate for the high-informed group, on the other hand, may reflect a weighted average of truly high-informed individuals and a subset of formerly low-informed individuals who upgraded their self-assessment after reading the article. Since we observe a larger effect size in the low-informed group, this contamination would likely inflate the effect size observed in the high-informed group.

As a result, the observed difference in treatment effects between low- and high-informed participants can be interpreted as a conservative estimate of the true moderation effect. The actual disparity in susceptibility to framing between the two groups may, in fact, be even greater than reported. Nonetheless, we acknowledge this limitation and recommend that future studies collect pre-treatment measures of informedness to more rigorously test moderation effects.

2.3 Principal component analysis

This section presents descriptive statistics from the principal component analysis (PCA) conducted on the six opinion-related questions asked for each event. The first principal component was used as a key outcome variable in our analysis to capture overall shifts in opinion.

Figure S7 displays the proportion of variance explained by each principal component across the seven events. In all cases, the first principal component accounts for the majority of the variance, indicating that it serves as a suitable summary measure for opinion-related responses. Figure S8 illustrates the contribution of each question to the first principal component, quantified by the cosine of the angle between each question vector and the first component. This helps to interpret the substantive meaning of the composite opinion measure used in the analysis.

2.4 Effects on individual opinion questions

To complement the main analysis using the first principal component of the opinion questions, we also report the treatment effects on each individual opinion item in Figure S9. This allows for a more granular interpretation of how specific aspects of opinion were influenced by the experimental treatments.

2.5 Robustness checks

In this section we perform robustness checks with two additional models similar to the regression model presented in Equation 1 with two additions. First, we cluster standard errors at the event-level, and second, we add random effects and slopes for events. Since both of these analyses are only relevant for the aggregate analysis where we analyze all events together, we only present the aggregate results in Figure S10. As you can see, the main results for the negative treatment is robust and remains statistically significant regardless of how we do the inference. However, the effect of the positive treatment which was very smaller in magnitude, looses its statical significance if we add clustering or random effects.

Furthermore, we explore the robustness of the results to the exclusion criteria for participants, based on their responses to the comprehension check questions. Figures S11 and S12 shows the results for the analysis including all responses, and only those with all

correct comprehension check responses. As you can see, the overall results are robust to how we filter the data.

2.6 Moderation by respondents' partisanship

We conducted an exploratory moderation analysis to examine whether treatment effects on the *feeling* outcome varied by participants' self-reported partisanship. Respondents were categorized into three main groups: Democrat ($n = 860$), Independent ($n = 547$), and Republican ($n = 457$). A fourth group of $n = 46$ respondents selected "Other" and manually entered diverse affiliations; given the small size and heterogeneity of this group, they were excluded from the analysis.

We estimated two linear models to examine heterogeneous treatment effects by partisanship. The first model excludes Event interactions and estimates only Treatment × Party effects, while the second includes a three-way interaction between Treatment, Party, and Event.

$$\begin{aligned} \text{Feeling}_i \;=\; & \beta_0 \;+\; \beta_1 \text{Treatment}_i \;+\; \beta_2 \text{Party}_i \;+\; \beta_3 (\text{Treatment} \times \text{Party})_i \\ & +\; \mathbf{X}_i \boldsymbol{\Gamma} \;+\; \epsilon_i \end{aligned} \tag{2}$$

$$\begin{aligned} \text{Feeling}_i \;=\; & \beta_0 \;+\; \beta_1 \text{Treatment}_i \;+\; \beta_2 \text{Party}_i \;+\; \beta_3 \text{Event}_i \;+\; \beta_4 (\text{Treatment} \times \text{Party})_i \\ & +\; \beta_5 (\text{Treatment} \times \text{Party} \times \text{Event})_i \;+\; \mathbf{X}_i \boldsymbol{\Gamma} \;+\; \epsilon_i \end{aligned} \tag{3}$$

where $\mathbf{X}_i$ represents the same demographic controls as in the main model. In this specification, the reference (baseline) categories were Neutral for Treatment, Independent for Party, and the ("Harris Picks Walz") for Event. All reported coefficients therefore capture differences relative to these baseline levels.

At the aggregate level (pooling across all events), we found no clear evidence of systematic moderation by partisanship (Figure S13, Panel A). However, when examining treatment effects separately by event (Panel B), clear patterns emerged especially for politically salient topics. For example, Democrats reacted more negatively to positive articles about the *Trump Assassination Attempt*, whereas Independents and Republicans responded more positively. Conversely, Republicans were largely indifferent to negative

coverage of *Harris Picks Walz*, while Democrats exhibited significantly more negative reactions.

Table S3 reports the full set of interaction coefficients for the *Treatment* × *Party* × *Event* terms. Among the significant results, Democrats showed stronger reactions to negative framing for relatively neutral or economic events (e.g., *Tax Bill*, *Boeing Grounding*, *Fed Rate Cut*), whereas positive coverage of right-leaning events such as the *Trump Assassination Attempt* elicited sharply negative reactions. Republicans, by contrast, exhibited fewer systematic differences, with only a marginally stronger response to the negatively framed *U.S.–Russia Prisoner Swap*.

These asymmetric, event-specific responses likely offset one another when averaged across all events, explaining the absence of an aggregate moderation effect. While exploratory, these results highlight the potential of the framework to detect subtle heterogeneity in framing effects across partisan groups—a promising direction for future research.

Table S3: Heterogeneous treatment effects on *feeling*: interaction coefficients for **Treatment × Party × Event**.

Event	Treatment	Party	Estimate	p	CI Low	CI High
Bipartisan Tax Bill Passed	Negative	Republican	−5.90	0.66	−32.29	20.50
Bipartisan Tax Bill Passed	Negative	Democrat	27.09	0.02	5.16	49.02
Bipartisan Tax Bill Passed	Positive	Republican	11.92	0.37	−14.38	38.22
Bipartisan Tax Bill Passed	Positive	Democrat	−2.90	0.80	−25.67	19.86
Boeing 737 Max 9 Grounding	Negative	Republican	−15.73	0.21	−40.47	9.01
Boeing 737 Max 9 Grounding	Negative	Democrat	36.83	0.00	14.95	58.71
Boeing 737 Max 9 Grounding	Positive	Republican	−12.42	0.35	−38.39	13.54
Boeing 737 Max 9 Grounding	Positive	Democrat	−10.81	0.36	−33.97	12.34
Fed Interest Rate Cut	Negative	Republican	−21.08	0.11	−47.00	4.84

Continued on next page

Event	Treatment	Party	Estimate	p	CI Low	CI High
Fed Interest Rate Cut	Negative	Democrat	28.12	0.02	5.43	50.82
Fed Interest Rate Cut	Positive	Republican	−12.00	0.40	−40.04	16.04
Fed Interest Rate Cut	Positive	Democrat	−16.98	0.16	−40.63	6.66
Imane Khelif's Olympic Journey	Negative	Republican	−1.75	0.89	−27.19	23.68
Imane Khelif's Olympic Journey	Negative	Democrat	13.41	0.23	−8.45	35.27
Imane Khelif's Olympic Journey	Positive	Republican	−8.42	0.53	−34.41	17.56
Imane Khelif's Olympic Journey	Positive	Democrat	−5.58	0.64	−28.74	17.57
Trump Assassination Attempt	Negative	Republican	−14.71	0.26	−40.11	10.68
Trump Assassination Attempt	Negative	Democrat	18.69	0.10	−3.68	41.05
Trump Assassination Attempt	Positive	Republican	−5.20	0.70	−31.92	21.53
Trump Assassination Attempt	Positive	Democrat	−43.64	0.00	−66.26	−21.02
US–Russia Prisoner Swap	Negative	Republican	−24.82	0.05	−50.14	0.49
US–Russia Prisoner Swap	Negative	Democrat	11.11	0.34	−11.75	33.98
US–Russia Prisoner Swap	Positive	Republican	−16.10	0.22	−41.87	9.67
US–Russia Prisoner Swap	Positive	Democrat	−10.60	0.37	−33.68	12.47

Notes: Entries are OLS interaction coefficients from the model with Treatment × Party × Event terms; baseline (reference) categories are Independents (Political Party), Neutral (Treatment) and Harris Picks Walz (Event). CI = 95% confidence interval.

References

[1] J. S. Wang, *et al.*, *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (2025), pp. 1–27.

Event	Positive	Neutral	Negative
Trump Assassination Attempt	This article frames the assassination attempt on Trump as a catalyst for political momentum and national unity, highlighting his resilience and the bipartisan support he received.	This article emphasizes the security failures and political turmoil resulting from the assassination attempt, focusing on the immediate response and implications for the upcoming Republican National Convention.	This article highlights the violence and political blame game following the assassination attempt, stressing the toxic political climate and its potential impact on the 2024 election.
Boeing 737 Max 9 Grounding	This article frames the incident as a testament to aviation safety and swift action, highlighting the professionalism of the crew and the industry's commitment to ensuring passenger safety.	This article emphasizes the technical issues and regulatory response, focusing on the discovery of loose parts and the FAA's urgent inspections to address safety concerns.	This article presents the incident as a significant setback for Boeing, highlighting the chaotic in-flight experience, financial repercussions, and ongoing scrutiny of the company's safety practices and transparency.
US-Russia Prisoner Swap	This article frames the prisoner swap as a historic diplomatic achievement for the Biden administration, emphasizing the joyous reunions and the power of international cooperation.	This article presents the prisoner swap as a significant diplomatic breakthrough involving multiple countries, highlighting the complexity and scale of the negotiations amidst geopolitical tensions.	This article criticizes the prisoner swap as a dangerous precedent that could encourage further hostage-taking by Russia, questioning the ethics and implications of the exchange.
Bipartisan Tax Bill Passes	This article frames the tax bill as a significant bipartisan achievement that provides financial relief to American families and businesses, emphasizing its potential to lift children out of poverty and enhance U.S. business competitiveness globally.	This article presents the tax bill as a rare bipartisan legislative success in a divided Congress, highlighting the compromise and challenges in passing the bill amidst political maneuvering and opposition from some lawmakers.	This article frames the tax bill as a controversial measure that has sparked criticism for disproportionately benefiting corporations and the wealthy, while failing to adequately support low-income families, amidst deep political divisions.
Imane Khelif's Olympic Journey	This article frames Imane Khelif as a national hero and symbol of resilience, focusing on her triumph over gender misconceptions and the overwhelming support from Algerian fans as she advances to the Olympic final.	This article emphasizes the broader cultural and political context of Khelif's journey, highlighting the support from Algerian communities in Paris and the challenges posed by gender scrutiny in sports.	This article presents a critical view of the controversy surrounding Khelif, focusing on the defamation and discrimination she faces due to gender misconceptions and the role of international organizations in perpetuating these issues.
Kamala Picks Tim Walz	This article frames the selection of Tim Walz as Kamala Harris's running mate as a positive and energizing choice for the Democratic base, highlighting his progressive achievements and appeal to a diverse electorate.	This article presents the selection of Tim Walz as a strategic and pragmatic move by Kamala Harris, emphasizing his moderate political background, military service, and potential to attract bipartisan support.	This article frames Tim Walz's selection as a controversial and risky decision, focusing on criticisms of his leftist policies, handling of past crises, and potential to be a liability for the Harris campaign.
Fed Interest Rate Cut	This article frames the Federal Reserve's interest rate cut as a positive and strategic move aimed at boosting the U.S. economy, achieving a 'soft landing,' and providing relief from high borrowing costs.	This article presents the rate cut as a cautious and preemptive measure by the Federal Reserve to balance inflation control with employment stability, while acknowledging internal dissent and the challenges of economic forecasting.	This article frames the rate cut as a controversial decision, sparking accusations of political motives and criticism from both political parties, with concerns about its timing and potential economic consequences.

Figure S4: Framing analysis of generated stimuli by GPT-4o, illustrating the coherent themes or frames that emerge for each event as a result of clustering sentences by tone.

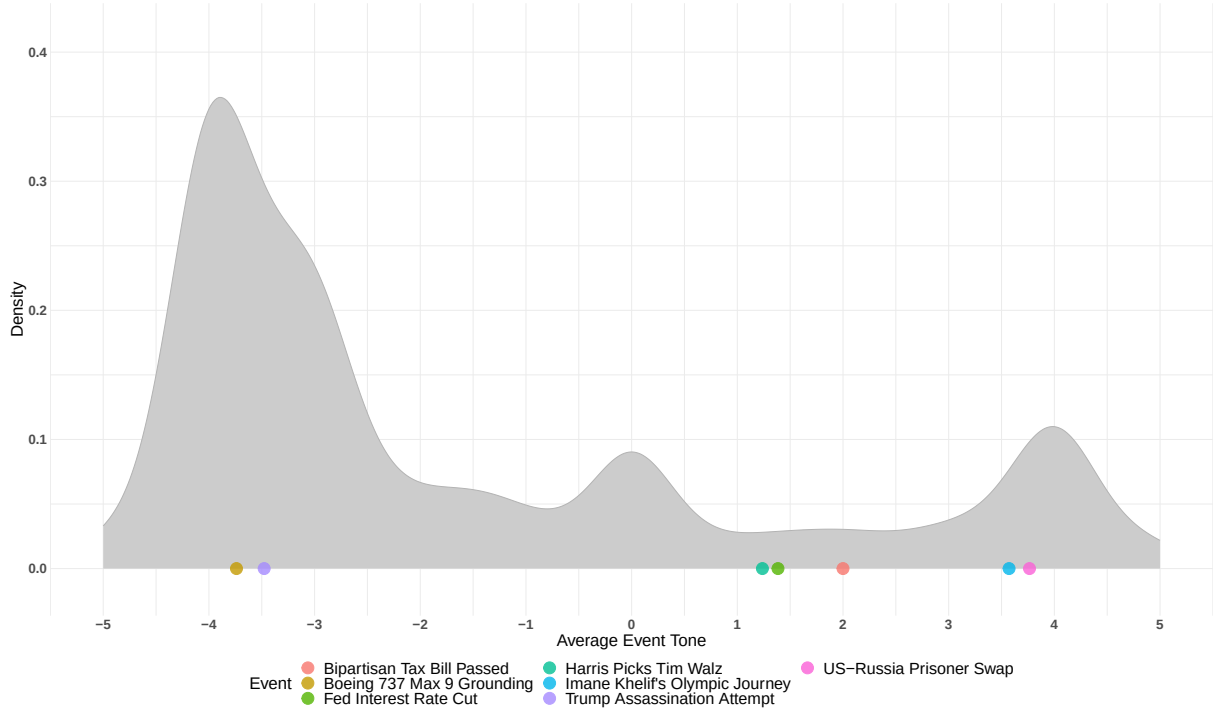


Figure S5: Distribution of coverage tone across all events from January to September 2024, highlighting a prevalence of negative tone. The figure also emphasizes the tone of events sampled for our experiment.

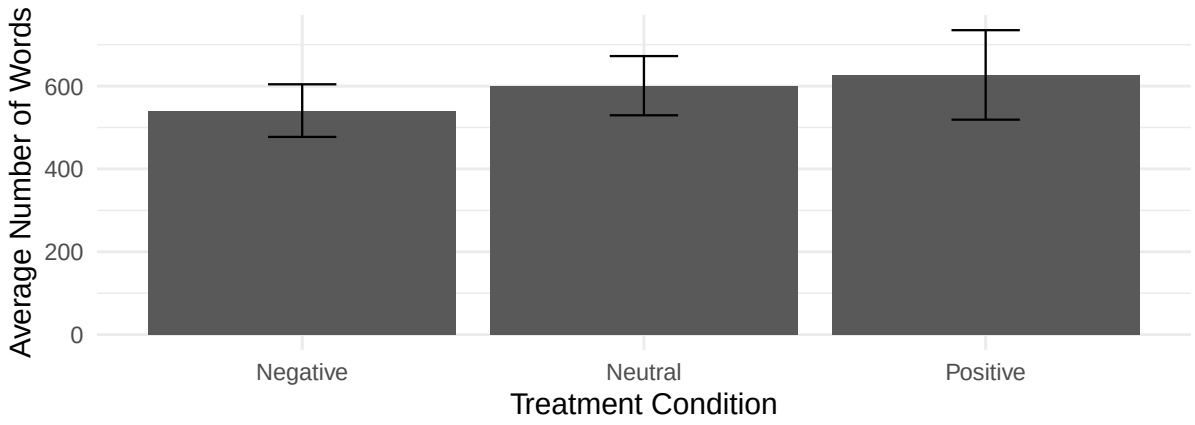


Figure S6: Mean word counts with 95% confidence intervals are shown for negative, neutral, and positive articles. While positive articles are slightly longer on average (610 words) than neutral (601) and negative (540) articles, these differences are not statistically significant across topics.

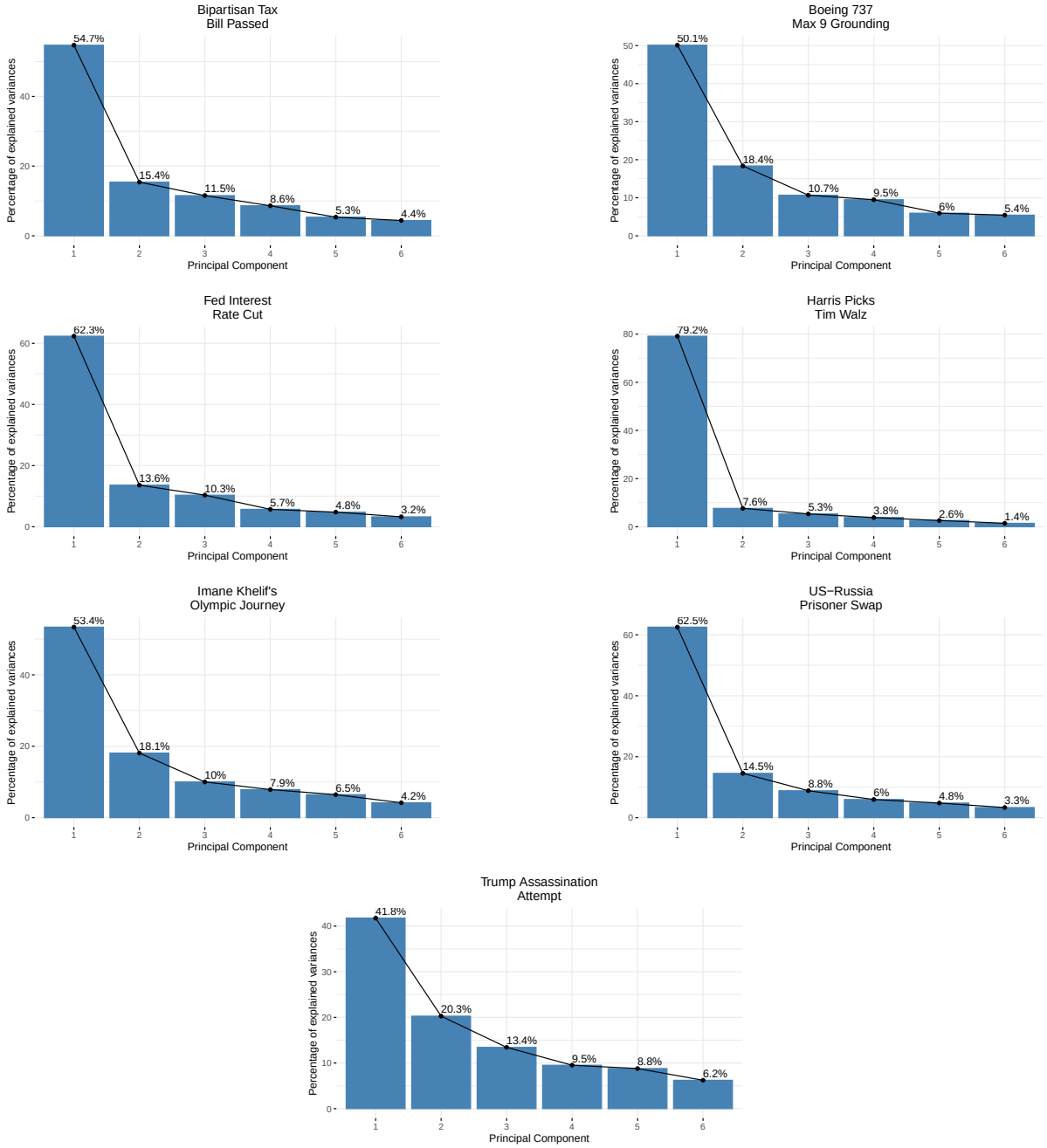


Figure S7: This figure displays the proportion of total variance explained by each principal component across the seven events. In all cases, the first principal component accounts for a substantial share of the variance, supporting its use as a summary measure of participants' opinions.

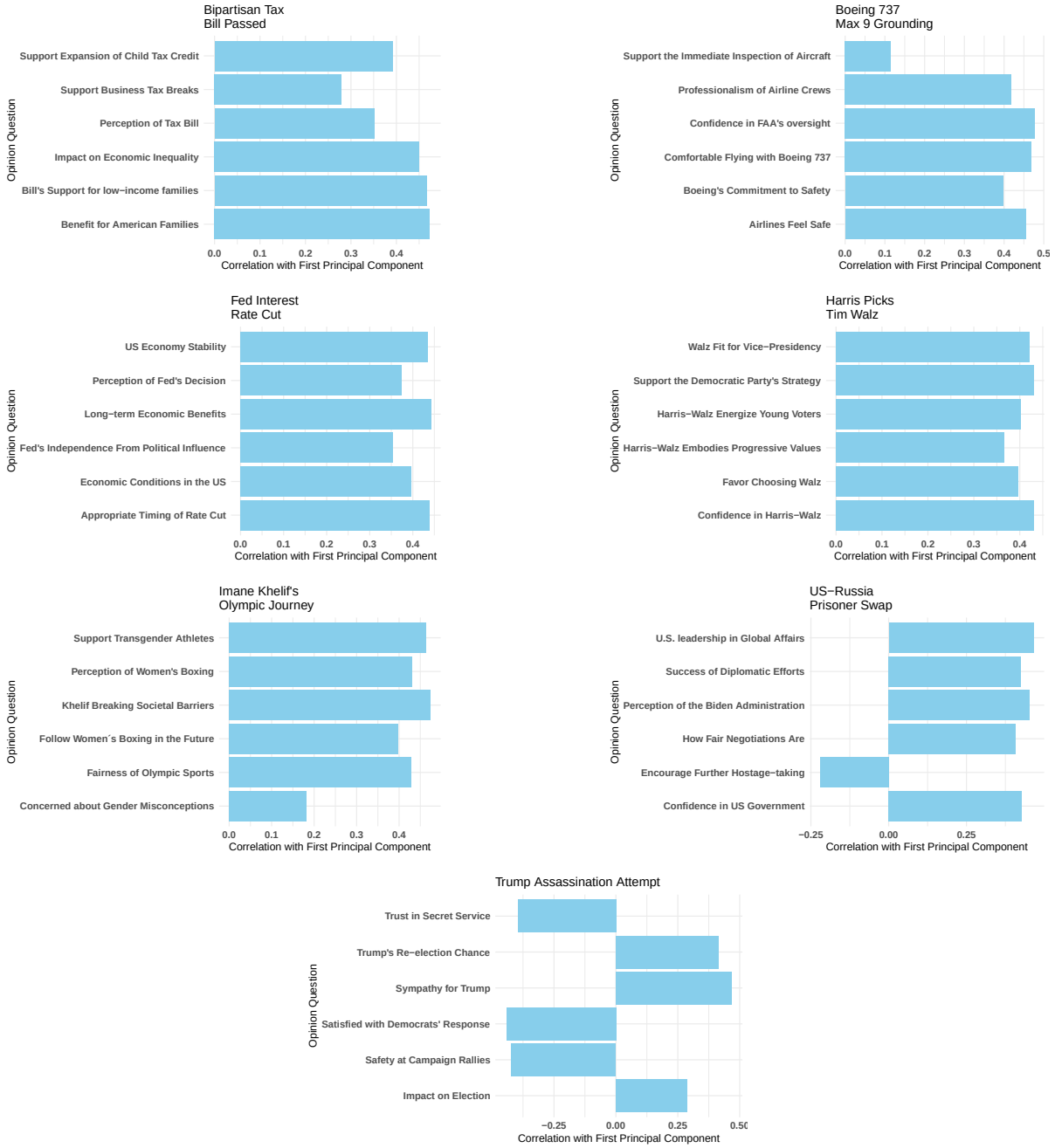


Figure S8: This figure shows the contribution of each opinion question to the first principal component, measured by the cosine of the angle between each question vector and the component. Higher values indicate greater influence of that question on the overall composite opinion score.

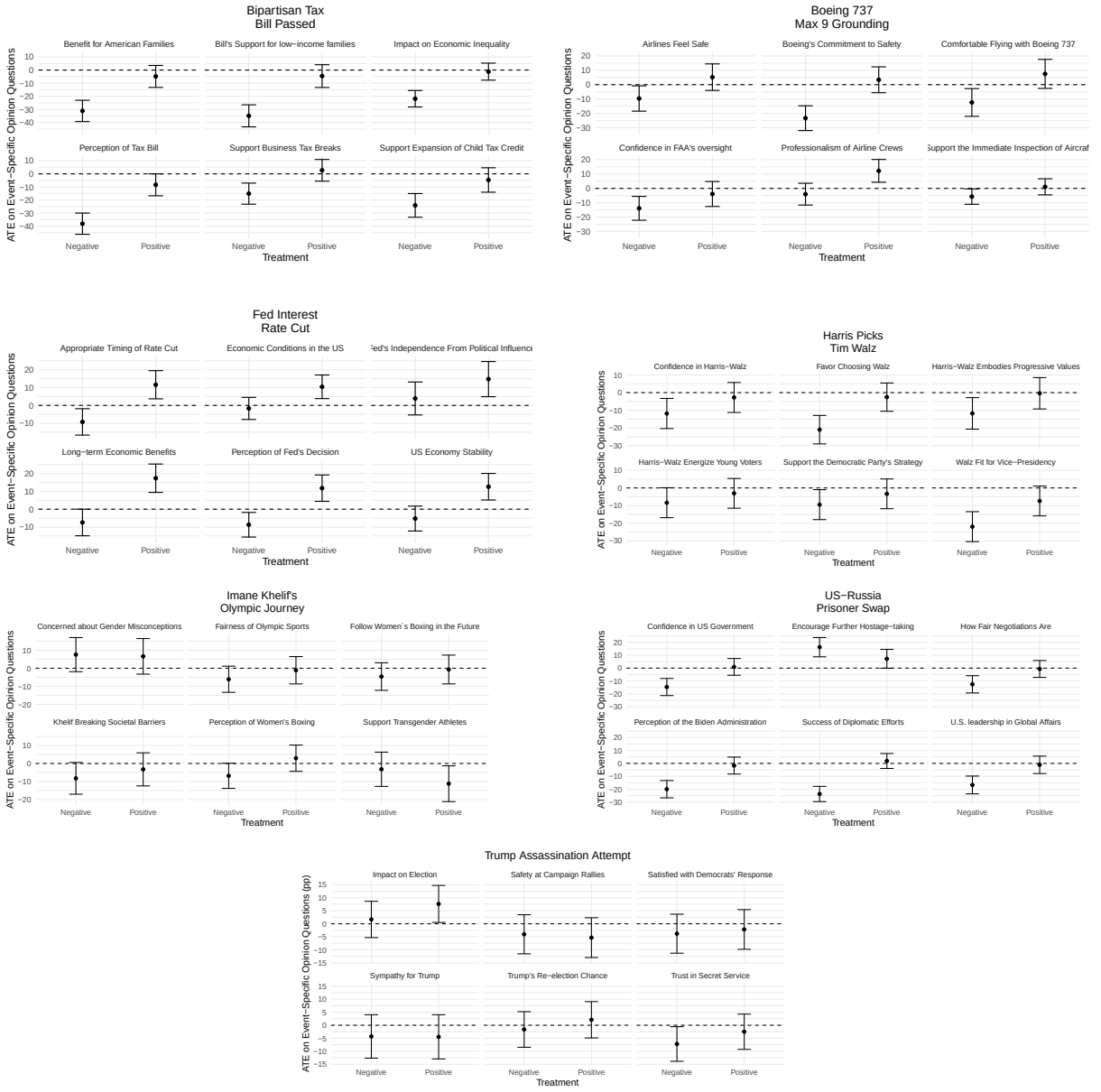


Figure S9: Treatment effects on individual opinion questions across all events.

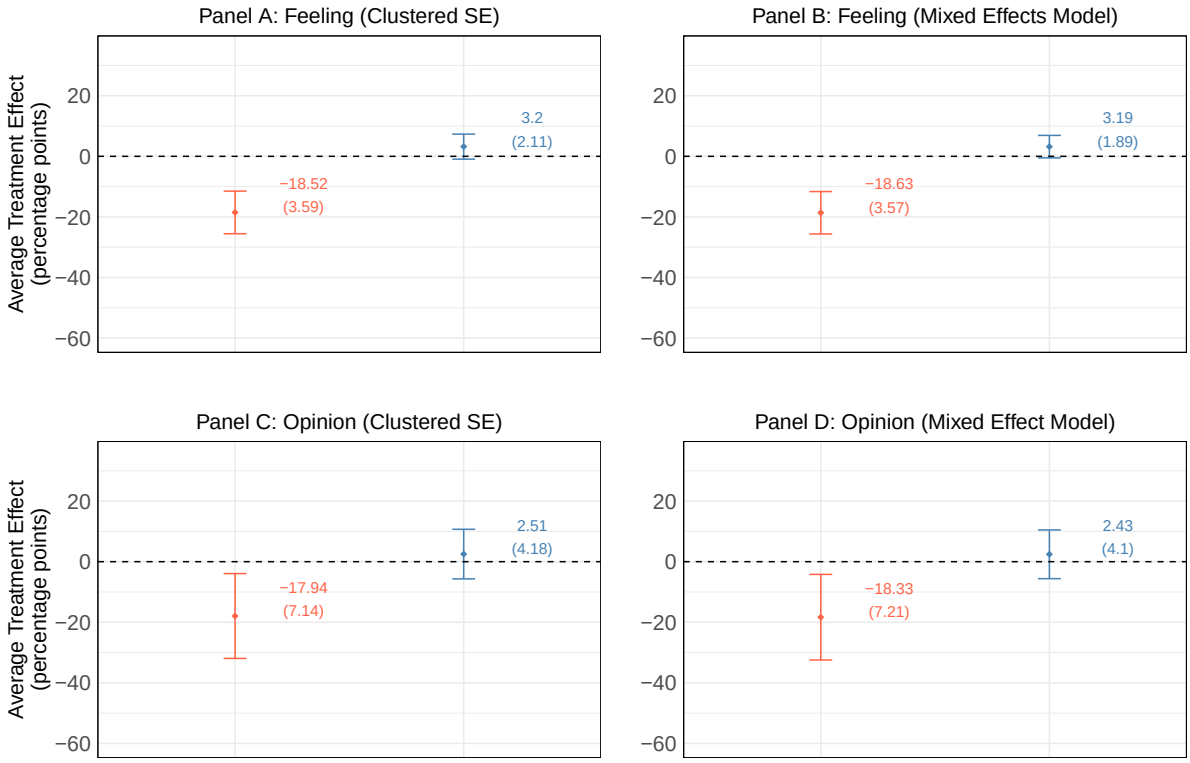


Figure S10: Robustness check for the main results with clustered standard errors and mixed effects models.

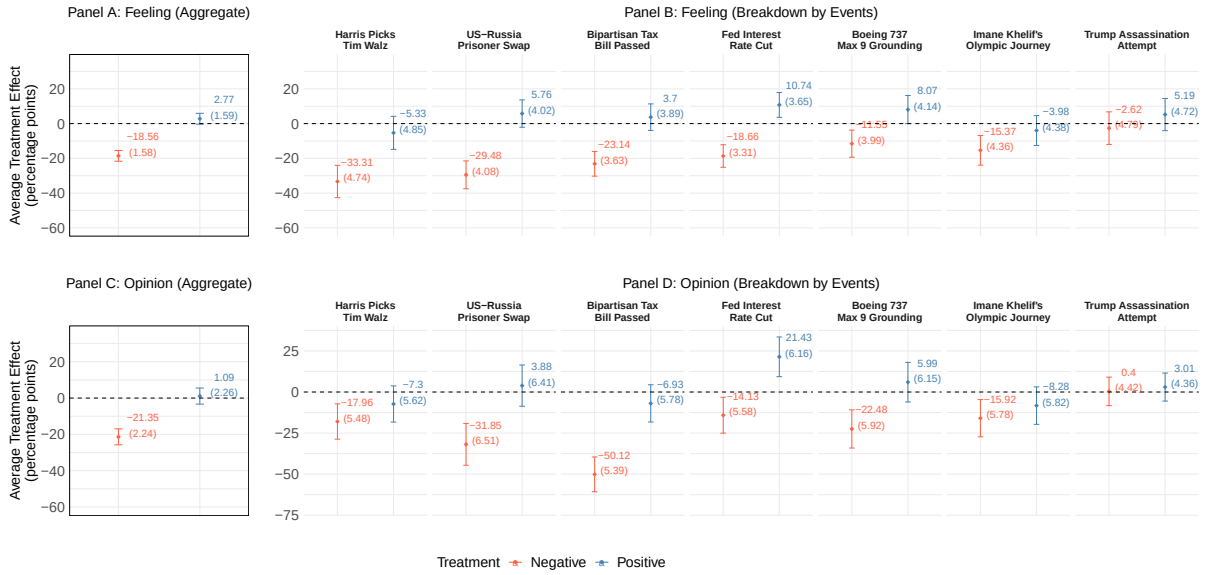


Figure S11: Replicating main results on feelings and opinions treatment effects (Figures 2 and 3 in the main manuscript), including all respondents, regardless of their answer to comprehension questions.

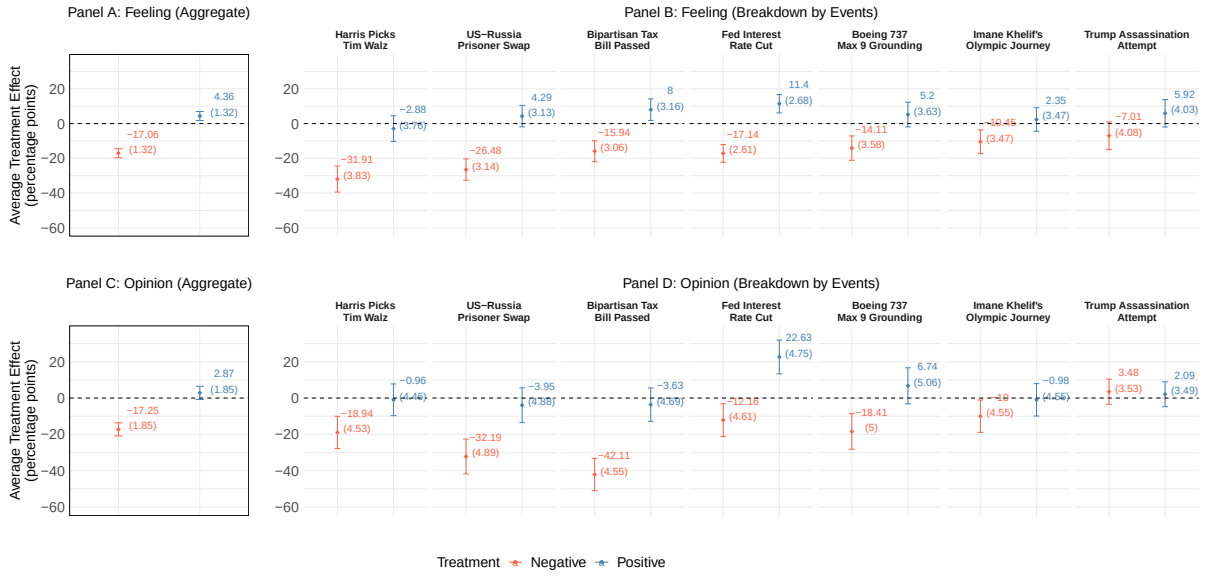


Figure S12: Replicating main results on feelings and opinions treatment effects (Figures 2 and 3 in the main manuscript), including only fully attentive respondents who answer all comprehension questions correctly.

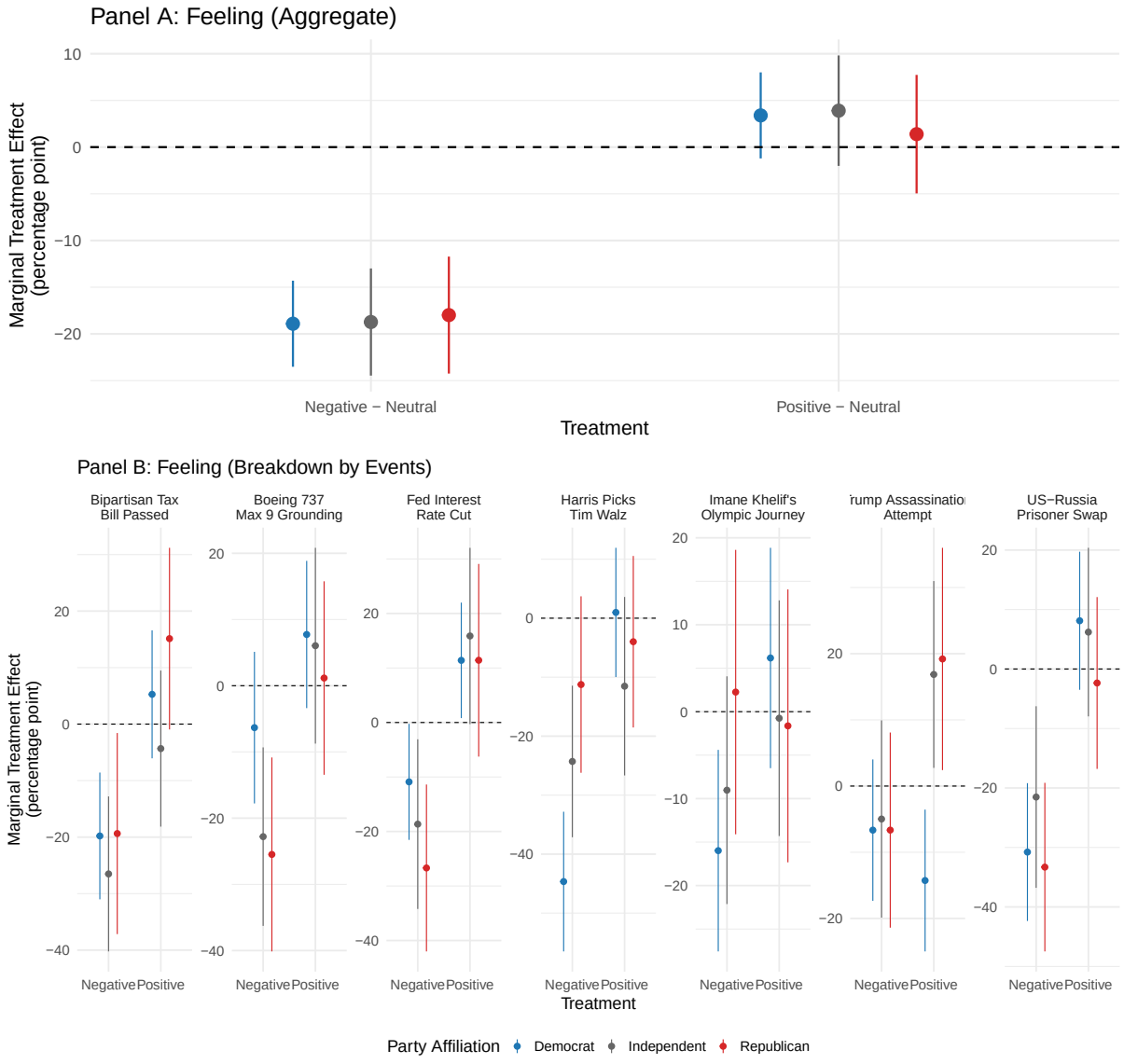


Figure S13: **Panel A** shows the aggregate treatment effects on feelings by partisanship (Independent, Republican, Democrat). **Panel B** breaks down the effects by individual events, revealing opposing patterns across politically charged topics.